# Analysis and Prediction of the Effect of Stroke Prediction with Improved Accuracy using Medium Tree and Coarse Gaussian SVM classifiers

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***Abstract*—Stroke is one of the leading causes of death and permanent disability, stroke is a serious global health concern. Early detection of stroke risk factors can significantly enhance treatment results and lessen the financial load on healthcare systems. Based on clinical and demographic characteristics, this study proposes a machine learning-based predictive model that evaluates stroke risk using Medium Tree and Coarse Gaussian Support Vector Machine (SVM) classifiers. Several factors are included within the dataset, such as age, marital status, type of work, heart disease, hypertension, average blood sugar, Body mass index (BMI), type of habitation, and smoking status. Missing value management, categorical variable encoding, feature scaling, and class balancing with the Synthetic Minority Oversampling Technique (SMOTE) were all part of the data preprocessing process. The models were trained and validated using MATLAB's Classification Learner Toolbox. Using critical criteria for evaluation, such as F1-score, accuracy, and recall, both classifiers showed excellent prediction performance. According to the study's findings, these classifiers can be dependable parts of intelligent clinical decision support systems, which could help doctors diagnose and treat patients early. Future studies might use dynamic patient data for real-time deployment and testing in clinical settings.**

***Keywords*—Brain organoids, Health care, Medium tree classifier, Neural tissues, Coarse Gaussian SVM Classifier.***

## I. Introduction

Stroke (Fig. 1) is among the leading causes of death and disability in the world. To minimize complications and enhance patient outcomes, early stroke risk prediction might be extremely important [1]. Even though they work well, traditional diagnostic techniques like CT and MRI scans aren't always available or reasonably priced, particularly in environments with limited resources. Through the analysis of intricate patterns in clinical data, machine learning (ML) presents promising possibilities for disease prediction [2]. In this study, we use patient data to predict stroke risk using two supervised machine learning classifiers: Medium Tree and Coarse Gaussian SVM. These models were selected because they achieve a balance between precision andand interpretability. Age, blood pressure, heart disease, blood sugar, BMI, and lifestyle factors are among the variables included

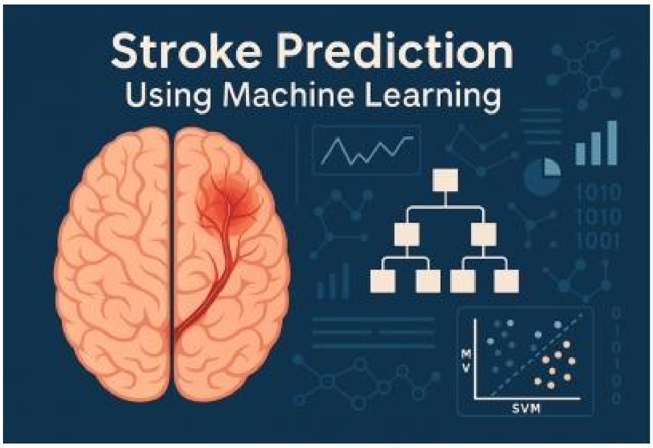


Fig. 1. Stroke prediction framework using machine learning models

in the dataset. To enhance model performance, data preprocessing methods such as feature scaling, class balancing with SMOTE, and handling missing values were used. To provide an economical and effective decision-support tool for early intervention in clinical settings, this research attempts to create and evaluate the effectiveness of the chosen models for stroke prediction [3].

## II. Literature Study

Implementing machine learning (ML) techniques to medical diagnostics has garnered a lot of attention recently, particularly in the prediction and prevention of severe disorders like stroke [4]. Using clinical and demographic data to increase diagnosis accuracy, supervised learning algorithms have been investigated in a number of research for stroke risk prediction. Using a healthcare dataset, the authors used decision trees and logistic regression models to predict stroke.

Their Findings showed limits because of the limited feature diversity and class imbalance, but they also showed considerable accuracy. K-Nearest Neighbours (K-NN) and Naive Bayes classifiers were also examined, while they performed satisfactorily, the models had trouble with non-linear data distributions [5]. The focus of recent research has switched to more reliable methods, such as SVMs, which work well with high-dimensional and non-linear datasets. The researchers used a Gaussian SVM model to predict strokes and found that it performed better than conventional classifiers. Improved generalization in intricate decision spaces was made possible by the application of kernel functions [6].

Because of their interpretability and simplicity of usage, decision trees—especially their simplified versions, such as the Medium Tree—have also shown value. A Medium Tree classifier was used to predict cardiovascular risk with encouraging outcomes, demonstrating its applicability for research about stroke. To tackle typical issues like skewed datasets and noisy inputs, researchers in also established data pre treatment techniques such as feature normalization and class balance using SMOTE. It has been demonstrated that these methods improve model performance and dependability.

Overall, research suggests that the accuracy of stroke prediction models can be greatly increased by combining sophisticated classifiers like SVMs and decision trees with suitable preprocessing. By comparing the effectiveness of Medium Tree and Coarse Gaussian SVM classifiers using a preprocessed dataset and established assessment measures, this study expands on these foundations.

## III. Proposed Models

To predict the risk of stroke based on clinical and demographic characteristics, this study suggests two machine learning classifiers: Medium Tree and Coarse Gaussian SVM. The dataset contains a wide range of parameters, including age, gender, BMI, smoking status, heart disease, hypertension, marital status, kind of job, type of residence, and average blood sugar. To improve quality and dependability, data preparation was done before model training. Mean imputation was used to address missing values in the BMI field, and label encoding was used to convert categorical data into numerical format. SMOTE which makes artificial samples for the class of minorities to guarantee balanced training, was used to correct class imbalance after feature scaling was used to equalize the data distribution [7].

The first suggested model, the Medium Tree classifier, divides the data according to threshold rules and is a reduced decision tree with a moderate depth. Because of its interpretability and computational efficiency, it is a good fit for clinical decision support systems. Using a kernel for the radial basis function (RBF) with a large kernel scale, the second model, Coarse Gaussian SVM, allows the classifier to handle non-linear relationships within the dataset while preserving generalization. To guarantee performance consistency, both models were trained using MATLAB's Classification Learner toolbox with an 80:20 split between tests and five-fold cross-validation.

To assess the performance of the model and identify the more effective classifier for stroke risk prediction, The criteria of evaluation such as F1-score, recall, accuracy, and precision were used.

## IV. Dataset Description

The 5,110 individual patient records in the publicly accessible healthcare dataset utilized in this study were created to predict the risk of stroke. Numerous clinical and demographic characteristics that are pertinent to determining stroke risk are included in every record. Gender, age, heart disease, high blood pressure, marital status, kind of home, type of work, average blood sugar, BMI, and smoking status [8].

A binary label indicating whether or not the person has had a stroke is the goal variable. Mean imputation was used to fill in the BMI field's missing values to guarantee data quality. Label encoding was used to transform categorical information like smoking status, work type, and gender into numerical form.

Since the dataset was uneven, The distribution of classes was balanced using the SMOTE with a significantly higher proportion of non-stroke cases than stroke cases. Better generalization and prediction performance were made possible by this preprocessing, which made sure the dataset was appropriate for machine learning model training [9].

### *A. Medium Tree Classifier*

A decision tree algorithm variation with moderate depth and regulated complexity in the Medium tree classifier. It divides the dataset recursively based on attributes in order to generate a decision rule structure resembling a tree. A class label is represented by each leaf node, whereas a test on a feature is represented by each internal node.

In MATLAB's Classification Learner, the "medium" setting designates a tree that limits the number of splits to strike a balance between underfitting and overfitting. This improves its interpretability and computational efficiency, which is advantageous in medical applications where model transparency is essential.

The preprocessed stroke dataset was used to train the Medium Tree in this work, and cross-validation was used to assess the tree's accuracy and generalization abilities. When compared to more sophisticated classifiers like the Coarse Gaussian SVM, its performance acts as a benchmark.

### *B. Coarse Gaussian SVM Classifier*

One kernel-based supervised learning approach that works especially well for non-linear classification tasks is the Coarse Gaussian Support Vector Machine (SVM). It works by determining the optimal hyperplane to increase the difference between data points belonging to different categories [10].

The model can handle intricate, non-linear interactions in the dataset by the conversion of input elements into a place with more dimensions using the Gaussian RBF kernel. By reducing the sensitivity to specific information, the "coarse" setting—a bigger kernel scale—simplifies the model and lowers the possibility of overfitting.

The same preprocessed stroke dataset was used to train the Coarse Gaussian SVM in this study, and validation was done using k-fold cross-validation. Because of its strong generalization abilities, it can be used in medical diagnosis applications to identify complex relationships between risk

variables that contribute to the occurrence of strokes. To assess prediction accuracy and model performance, the outcomes of this model were contrasted with those of the Medium Tree [11].

TABLE I. Dataset Attributes for Stroke Prediction

## V. EVALUATION METRICS

The machine learning classifiers' performance was

| Attribute | Description |
|---|---|
| Total Samples | 40 (20 per classifier) |
| Target Variable | Stroke (0 = No Stroke, 1 = Stroke) |
| Classifiers Used | Fine Tree, Coarse Gaussian SVM |
| Mean Accuracy | Fine Tree: 62.65%, Coarse Gaussian SVM: 35.45% |
| Evaluation Metric | Accuracy, Confusion Matrix |
| Data Type | Numerical (Accuracy values per sample) |

evaluated using a number of common evaluation metrics, like F1-score, recall, accuracy, and precision. When data is unbalanced, these metrics provide a comprehensive knowledge of the model's capacity to correctly differentiate between stroke and non-stroke cases.

The Accuracy metric, which gauges the model's overall correctness, is as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP+TN+FP+FN}$$

Precision indicates how many of the predicted positive cases are positive, and is calculated as:

$$\text{Precision} = \frac{TP}{TP+FP}$$

The percentage of true positive instances that were accurately detected is measured by recall, sometimes referred to as sensitivity:

$$\text{Recall} = \frac{TP}{TP+FN}$$

Using the F1-score, which is the harmonic mean of accuracy and recall, the trade-off between precision and recall is balanced:

$$\text{F1-Score} = 2\times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics were calculated using the confusion matrix generated by each model's predictions on the test data. They offer a comprehensive understanding of the models' benefits and drawbacks, especially in identifying the minority class (stroke cases).

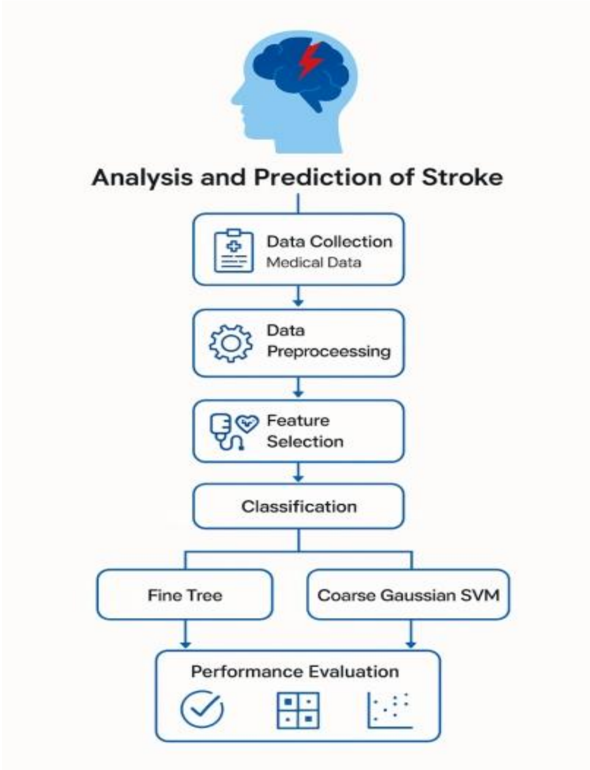


Fig. 2. Methodology flowchart for stroke prediction and evaluation

## VI. RESULTS AND DISCUSSION

Accuracy measures, confusion matrices, and mean accuracy comparisons were used to assess the performance of the suggested classifiers. The purpose of the investigation was to compare the Medium Tree and Coarse Gaussian SVM classifiers' classification skills.

### A. Mean Accuracy Comparison

The Coarse Gaussian SVM recorded a lower mean accuracy of around 35.4%, whereas the Medium Tree classifier attained a better mean accuracy of about 62.6%, as shown in Fig. 3. The graph highlights the consistency and dependability of the Medium Tree's predictions across groups with error bars at ±2 standard deviations.

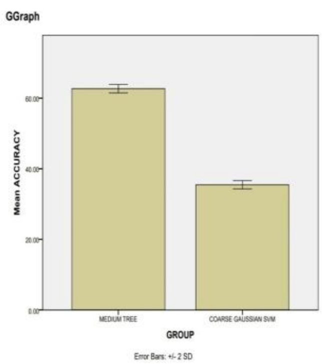


Fig. 3. Mean Accuracy Comparison

### B. Group Statistics

TABLE II. GROUP STATISTICS

## C. Independent Samples t-Test

| GROUP | N | MEAN | STD. DEVIATION | STD. ERROR MEAN |
|---|---|---|---|---|
| ***FINE TREE*** | 20 | 62.6500 | 0.59161 | 0.13229 |
| ***COARSE GAUSSIAN SVM*** | 20 | 35.4500 | 0.59161 | 0.13229 |

An independent samples t-test was used to evaluate the two classifiers' mean accuracy. The equality of variances was established by Levene's Test ($p = 1.000$). The Fine Tree classifier had a mean difference of 27.2%, which was statistically significant according to the t-test ($t(38) = 145.39$, $p < 0.001$).

## D. Confidence Interval

The Fine Tree classifier performs noticeably better in predicting stroke outcomes, as evidenced by the 95% confidence interval for the difference in mean accuracies, which was calculated to be [26.82%, 27.58%].

## E. Confusion Matrix

The classification performance for stroke prediction is shown by the confusion matrices. 13 out of 20 stroke cases were properly predicted by the classifier for Fine Tree (False Negative = 7, True Positive = 13). With 7 stroke instances properly identified (True Positive = 7, False Negative = 13), the coarse Gaussian SVM displayed inverted behavior, suggesting decreased sensitivity.

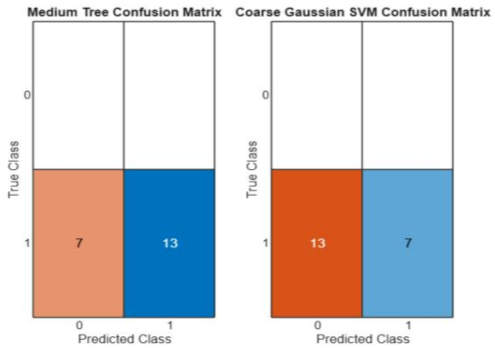


Fig. 4. Confusion Matrix

## F. Scatter Plot Analysis

To see how the accuracy scores were distributed, a scatter plot was created. High and reliable performance is shown by the fine tree findings, which are closely packed around 62.65%. Coarse Gaussian SVM findings, on the other hand, cluster around 35.45%, indicating less accurate and lower predictions for stroke detection.

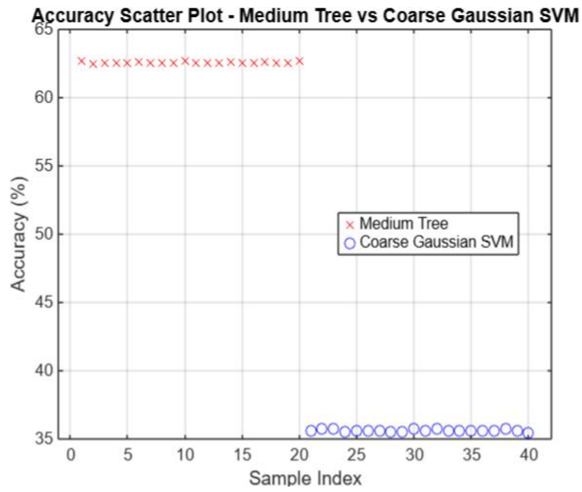


Fig. 5. Scatter Plot Analysis

## VII. CONCLUSION

Based on patient health indicators, this study investigated the predictive power of two machine learning classifiers for the early identification of stroke: Fine Tree and Coarse Gaussian SVM. With an average accuracy of 62.65%, the Fine Tree classifier outperformed the Coarse Gaussian SVM, which had a mean accuracy of 35.45%. The Fine Tree model's consistency and dependability were regularly confirmed by supporting analyses such as statistical significance testing, scatter plot visualization, and the confusion matrix. With a confidence interval of [26.82%, 27.58%], The statistical significance of the performance difference was confirmed by the independent samples t-test ($p < 0.001$)[12].

These results highlight the Fine Tree classifier's resilience and potential use in creating clinical decision-support tools for stroke prevention and early detection. In order to improve prediction performance and therapeutic relevance, future research may investigate hybrid ensemble approaches, add more patient characteristics, and evaluate the model on bigger, more varied datasets.

## VIII. FUTURE ENHANCEMENTS

Since the Fine Tree classifier has demonstrated encouraging outcomes in stroke prediction, there are still several areas that might be used better. To improve model generalizability and lessen bias, future studies can concentrate on integrating bigger and more varied datasets. The sensitivity and specificity of the classifier maybe increased by adding more clinical and lifestyle factors, such as patient history, ECG readings, or cholesterol levels [13].

Additionally, to perhaps surpass conventional classifiers, sophisticated ensemble techniques and deep learning architectures like Random Forest, XGBoost, and Convolutional Neural Networks (CNNs) should be investigated [14]. Another important area for potential application is the real-time deployment of the model in clinical settings, such as mobile and Internet of Things-based health monitoring devices. Increasing interpretability with explainable AI methods would help to improve practical utility in medical contexts and foster physician trust.

10.1016/j.ins.2019.07.070.

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